

Field Experiments

Methods and Applications

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- 1 Introduction to field experiments
- 2 Methods
- 3 Failures, successes and applied tips in the field

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Treatment Effects and Counterfactual

- We start from the assumption that the aim of the researcher is to estimate a causal effect of some action (treatment effect)
- The fundamental difficulty that arises is that either the action is taken or it is not (counterfactual)
- Key question: how to construct a proper counterfactual?
- Several approaches (Harrison & List 2004):
 - Randomization, (controlled experiments, e.g., laboratory, field)
 - Quasi-Randomization (e.g., quasi-experiment)
 - Propensity score matching
 - Instrumental variables

Trends in Econ Publications

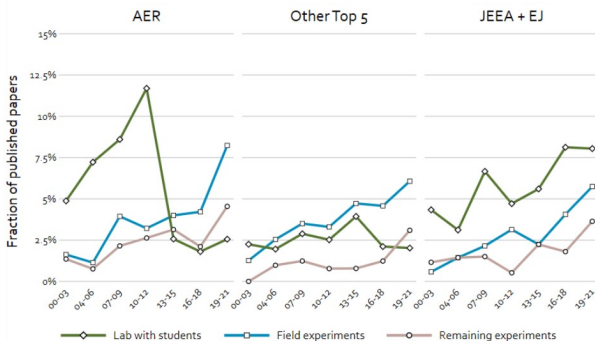


Figure A1 – Fraction published experimental articles per year

Figure 1: from Reubens et al. 2022

Definitions

- *Conventional Lab Experiment*: student pools, abstract framing, imposed set of rules
- *Artefactual field experiment*: same as Conventional Lab but with non-student pools
- *Framed field experiment*: same as Artefactual but with field context in either the task or info subjects can use
- *Natural field experiment*: same as Framed but environment is where subjects naturally undertake tasks, and they do not know to be in an experiment (naturally occurring experiments)

Note: some experiments may share features of more categories (see case studies)

Artifactual Field Experiments

Abstract task or commodity, Non-standard populations

Some examples:

- Standard experiments (PD, UG) run in peasant societies (tribes) (see Henrich 2000)
- Asset markets with subjects that have experience in trading operations (see Smith, Suchanek and Williams 1988)
- Elderly subjects
- Managers

Framed Field Experiments

No abstract tasks or commodity, and participants may have field experience or information

Example w. no abstract commodity: Bohm's experiments on public goods evaluation

- eliciting willingness to pay (WTP) for cable TV in groups (replicating a public good)
- if group WTP $>$ min. threshold imposed, then group has access to cable TV

Natural Field Experiments

Natural environment, and subjects are not informed they are part of an experiment

- Reducing energy consumption using smart meters (Allcott 2011)

- ### ③ Failures, successes and applied tips in the field

Measurement & Identification

Objective: evaluate the effect of an intervention (policy) by constructing a proper counterfactual

- Y_i is our outcome of interest measured on person i
- Potential Outcome: if i receives the treatment we observe $Y_i(1)$, otherwise $Y_i(0)$
- We measure out *Treatment effect* using the **mean difference estimator**:

$$\tau_i = Y_i(1) - Y_i(0)$$

- Average Treatment Effect:

$$E[Y_i(1)] - E[Y_i(0)] = \frac{1}{N} \sum_i \tau_i$$

Randomization

- We wish to study the budgetary consequences of having women, rather than men head Indian village councils, which govern rural areas in India (Chattopadhyay & Duflo 2004)

<i>Village i</i>	$Y_i(0)$ <i>Budget share if village head is male</i>	$Y_i(1)$ <i>Budget share if village head is female</i>	τ_i <i>Treatment effect</i>
Village 1	10	15	5
Village 2	15	15	0
Village 3	20	30	10
Village 4	20	15	-5
Village 5	10	20	10
Village 6	15	15	0
Village 7	15	30	15
Average	15	20	5

$$E[Y_i(1)] - E[Y_i(0)] = 20 - 15 = 5$$

Randomization

- Missing counterfactual problem: we often do not observe both states

$$Y_i = Y_i(0)(1 - d) + Y_i(1)d;$$

- Randomization as solution

<i>Village i</i>	$Y_i(0)$ <i>Budget share if village head is male</i>	$Y_i(1)$ <i>Budget share if village head is female</i>	τ_i <i>Treatment effect</i>
Village 1	?	15	?
Village 2	15	?	?
Village 3	20	?	?
Village 4	20	?	?
Village 5	10	?	?
Village 6	15	?	?
Village 7	?	30	?
Estimated average based on observed data	16	22.5	6.5

Randomization

If the treatment is administered *randomly* (all villages have equal chances to be treated) then, the treatment group is a random sample of all villages, and therefore the expected potential outcomes among villages in the treatment group are identical to the average potential outcomes among all villages.

Formally, let us indicate the treatment group with $T = 1$, $T = 0$ otherwise:

$$E[Y_i(1)|T = 1] = E[Y_i(1)] \quad (1)$$

$$E[Y_i(1)|T = 0] = E[Y_i(1)] \quad (2)$$

hence

$$E[Y_i(1)|T = 1] = E[Y_i(1)|T = 0] \quad (3)$$

similarly

$$E[Y_i(0)|T = 0] = E[Y_i(0)|T = 1] = E[Y_i(0)] \quad (4)$$

We can rewrite the ATE as:

$$E[Y_i(1)|T = 1] - E[Y_i(0)|T = 0] \quad (5)$$

Randomization procedure: an example

- All units have equal chances to end up being treated
- *Simple random assignment*: Flip a coin; problem: small chance to end up with $N/2$ obs. in the treatment
- *Complete random assignment*: set m of N subjects to be included in the treatment with equal probability
 - Flip a coin until you get m units in the treatment
 - Enumerate all subjects, then compute all possible ways to group m subjects, and randomly select one
 - Rely on a statistical software

```
1 set.seed(1234567) # set random seed
2 N <- 7           # number of units
3 m <- 2           # number of treated units
4 sel <- rnorm(N)   # N standard Normal deviates
5 Tr <- order(sel)  # ordering permutation which orders the deviates
6 Tr[1:m]          # the first m indexes are treated units
7
```

What if no randomization?

Assume that the treatment is given not at random, but it depends, for example, on some village characteristics (selection bias)

ATE becomes (by adding and subtracting $E[Y_i(0)|T = 1]$) :

$$E[Y_i(1)|T = 1] - E[Y_i(0)|T = 1] + E[Y_i(0)|T = 1] - E[Y_i(0)|T = 0] \quad (6)$$

- the red component is the ATE among the treated (ATT)
- the blue component is the selection bias
- under randomization the selection bias is zero
- when treatment is not given at random, the bias is different than zero

Two Core assumptions

Excludability

- To be able to isolate the effect of the manipulated variable from other potential explanatory variables
- Potential outcomes respond to treatment solely, not to any direct or indirect consequences of randomization
- Example: administering a pre-experimental questionnaire to treated groups and not to control groups

Non-Interference:

- Administering of treatment does not affect potential outcomes of non-treated subjects
- Example: subjects in treatment interact with subjects in control

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Blocked random assignment

Cluster random assignment

Using Covariates in Experimental Design and Analysis

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Blocked randomization

- Randomization of treatment can happen also depending on some factor (gender, geography, working class) that form subgroups of the population
- Block random assignment refers to a procedure whereby subjects are partitioned into subgroups (called blocks or strata), and complete random assignment occurs within each block
- It is important to make sure that treatment assignment is yet random within each subgroups

Blocked randomization: pros and cons

Pros:

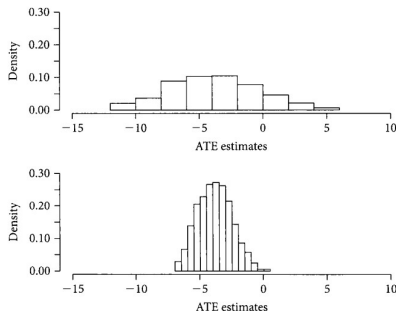
- Practical or ethical imperatives dictate the number of treatments that may be assigned to certain groups of subjects
- Useful in case of heterogeneous analysis
- Block randomization can increase precision of ATE estimates

Cons:

- No downsizes

$$ATE = \sum_j \frac{N_j}{N} ATE_j \quad SD(ATE) = \sqrt{\sum_j \left(\frac{N_j}{N}\right)^2 \times SE^2(ATE_j)}$$

Sampling distribution under complete randomization (above);
sampling distribution under blocked randomization (below)



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Cluster random assignment

- To this point, we have considered only experiments in which each of the subjects is individually assigned to the treatment or control group
- This is not always feasible (e.g., advertisement targeting geographic areas, policy affecting whole schools)
- Cluster assignment: population N divided into k clusters, and treatment assignment is done at the cluster level

Cluster random assignment

- **Warning:** ATE variability can be dramatically higher under cluster random assignment than complete random assignment
- The variability penalty associated with clustering depends on the variability of the cluster-level means
- The higher the heterogeneity *across* clusters, the more variable is the ATE estimator

Not-Ideal scenario

Hypothetical schedule of potential outcomes for 12 classrooms in four schools—high sampling variability

School	Classroom	Classroom-level potential outcomes			Cluster-level mean potential outcomes	
		$Y_i(0)$	$Y_i(1)$		$Y_i(0)$	$Y_i(1)$
A	A-1	0	4	}	1	5
A	A-2	1	5			
A	A-3	2	6			
B	B-1	2	6	}	3	7
B	B-2	3	7			
B	B-3	4	8			
C	C-1	3	7	}	4	8
C	C-2	4	8			
C	C-3	5	9			
D	D-1	7	11	}	8	12
D	D-2	8	12			
D	D-3	9	13			

Ideal scenario

Hypothetical schedule of potential outcomes for 12 classrooms in four schools—low sampling variability

School	Classroom	Classroom-level potential outcomes			Cluster-level mean potential outcomes	
		$Y_i(0)$	$Y_i(1)$		$Y_i(0)$	$Y_i(1)$
A	A-1	0	4	}	4	8
A	A-2	3	7			
A	A-3	9	13			
B	B-1	2	6	}	4	8
B	B-2	3	7			
B	B-3	7	11			
C	C-1	1	5	}	3.3	7.3
C	C-2	4	8			
C	C-3	5	9			
D	D-1	4	8	}	4.7	8.7
D	D-2	8	12			
D	D-3	2	6			

Cluster randomization - tips for the field

- Cluster randomization may lead to higher variability of ATE estimates
- Solution: increase number of clusters rather than obs. within a cluster

$$SE(ATE) = \sqrt{\frac{1}{k-1} \left(\frac{m \cdot \text{Var}(\bar{Y}_j(0))}{N-m} + \frac{(N-m) \cdot \text{Var}(\bar{Y}_j(1))}{m} + 2\text{Cov}(\bar{Y}_j(0), \bar{Y}_j(1)) \right)}$$

where m is the number of subj. in the treatment and k number of clusters

- Note: if n. subjects across clusters is unequal $\rightarrow$ ATE estimator is biased
- Increasing k is useful to reduce bias

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Using Covariates

- The main advantage of randomization: estimating ATE regardless of whether we account for other variables that can affect the outcome of interest
- Issues due to *omitted variables* are addressed by random assignment, as long as randomization is independent on all observed and unobserved variables
- However, to increase ATE estimates precision, we can include *covariates* in the design and analysis
 - Any variable that can be collected before or after experimental intervention, and gives background information of sample (e.g., age, gender, work conditions)
 - **Note:** they can vary over time, yet treatment intervention should not affect them, especially if collected after (Excludability principle)

Using Covariates

Three ways to achieve this goal:

- Rescaling outcomes
- Regression models ←
- Block randomization ←

ATE as regression model

ATE equation as a regression model:

$$Y_i = Y_i(0)(1 - d_i) + Y_i(1)d_i = Y_i(0) + (Y_i(1) - Y_i(0))d_i$$

Which is equivalent to:

$$Y_i = \mu_{Y(0)} + [\mu_{Y(1)} - \mu_{Y(0)}]d_i + u_i$$

or equivalently:

$$Y_i = a + bd_i + u_i$$

where b is the ATE

Regression model with covariates

If we include the covariates X_i

$$Y_i = a + bd_i + cX_i + (u_i - cX_i) =$$

$$a + bd_i + cX_i + u_i^*$$

- c has no causal interpretation, as covariates play no causal role
- the inclusion of X_i reduces the amount of unexplained variation in Y_i
- X_i are thought to predict outcomes, not to affect them

Block Randomization using covariates

Block what you can, randomize what you cannot

- Assume that you have identified some covariates that may predict potential outcomes
- You can form blocks based on these covariates (age, gender, income classes)
- Whenever possible, it is best to randomize within blocks to ensure highest precision in ATE estimates
- Overcome the possibility of uneven distribution across conditions with respect to important factors

Discussion on using covariates

- Pros: Improve precision
- Cons: If sample small ($N < 20$), covariates inclusion can lead to biased ATE estimates
- Inclusion of covariates should follow theory, pilots or any reasonable motivation
- Planning ahead which covariates to include beforehand to prevent p-hacking (PAP)

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5 Reasons that lead to failing in the field

- Inappropriate research design
- Technical design flaws
- Partner organization challenges
- Survey and measurement execution problems
- Low participation rate

1. Inappropriate research design

- Context matters: Research should be conducted in settings where the problem exists and the intervention is relevant.
- Timing issues: external events like politics or weather can disrupt research
- Feasibility: Ensure infrastructure and resources are available
- Knowing when to stop: do not be afraid to drop the study

Some tips:

- Conduct pre-experimental research to gather preliminary info from the field
- Do everything that can be informing for your research design (questionnaires, focus groups, co-design)
- Conduct power analyses before jumping in the field

An example: Guido et al. 2024

Problem. Switching from mobile to Wi-Fi data consumption (offloading)

Before going to the field and design interventions, get familiar with the problem at hand!

Preliminary questionnaire: knowing whether people know about offloading and what interventions would be most effective

- more than 50% of subjects did not know about the offloading problem → Inform them!
- Mobile consumption correlates with impatience → Improve self control (commitments)

Co-design sessions: group-up approach to designing experiments

Did you know that mobile networks produce up to 4 times more CO₂ than Wi-Fi networks?

According to your declaration of **Mobile** data consumption, you will produce:

2290g of CO₂/per month

If you had used **Wi-Fi** only with the same amount of consumed data, you will produce only:

536g of CO₂/per month

By **switching** to **Wi-Fi** network, your **CO₂ emissions will be reduced** by a quantity equivalent to:

32h electricity heating	
219 Km in High speed train	
2928h of LED bulb on	

As a reminder, here is your breakdown of last week's data consumption:

38% via Wi-Fi

62% via Mobile

Counter-Example: Romaniuc, Guido et al. 2024

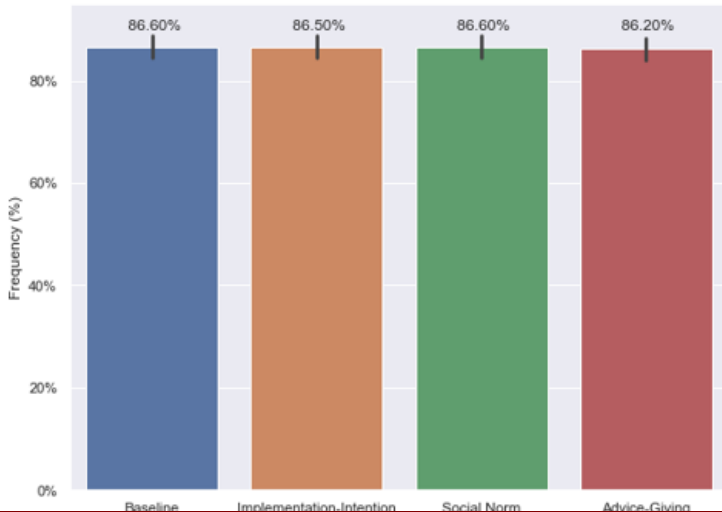
Problem. Nudging voter participation among young adults

Co-design sessions



cont'd

Yet, no pre-experimental hints on baseline participation rates



2. Technical Design Flaws

- **Survey Design Errors:** Question wording, response scales and question order
- **Survey Length and Relevance:** Tendency to add unnecessary questions
- **Statistical Power and Sample Size**
- **Randomization Protocols**

3. Partner organization

- Researchers often collaborate with partner organizations to implement development programs and gather relevant data.
- It is important to establish a *trust* relation with them
- Show the advantages of using experimental methods, do not use complex and cryptic jargon

4. Survey and Measurement problems

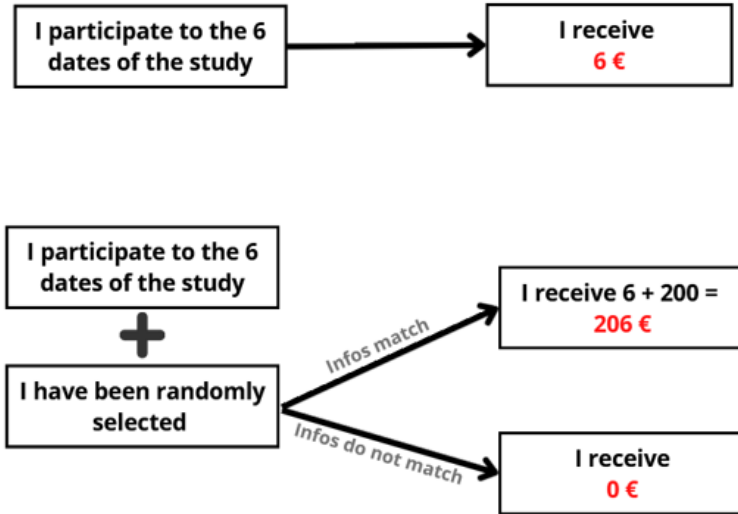
- Surveyor Misconduct: when they do not follow protocols
- Tracking respondents: if your study is longitudinal, make sure you can track your subjects (technologies?)
- Measurement tools: prioritize objective measures than human-reported ones (technologies, again?)

Some tips:

- Continuous monitoring: build a process for monitoring surveyor (meetings, auditing)
- No objective data? Use incentives to ensure truthful report

Example of incentive for truth (Guido et al. 2024)

- Use of third-party app to monitor data consumption
- Ask subjects to report their weekly consumption on our questionnaire every week
- How to reduce misreporting? **Random check at the end for k participants**
- They send screenshots of the app showing *real* consumption and we compare with declarations
- If no misreporting or errors, they get 200€, nothing otherwise



5. Low participation rates

- Distinguish between *during intake* and *after treatment assignment*
- *During intake*: less participants than expected
 - sampling, hard-to-reach participants
 - wrong incentives
 - trust in the researchers or organizations involved
- *After treatment assignment*: potential bias in estimates
 - treatment makes harder to stay in the study?

Example: Loan Repayment in Nigeria (Guido & Romaniuc, in progress)

Problem. Measuring external validity of reciprocity measure (trust game $\iff$ loan repayment)

- Bank provides contacts
- Researchers (w/ local RA) send invitation emails
- Response rate: 5% !!!
- Solutions: Involvement of local university + stricter protocol (phone calls with pre-defined script)

Thank You